Course Code	21EIC504J	Course Name	REAL TIME EMBEDDED SYSTEMS	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	0	2	4

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*Electronics and Instrumentation Engineering*	**Data Book / Codes / Standards**	*Nil*		

Course Learning Rationale (CLR):	***The purpose of learning this course is to:***
CLR-1:	*impart the knowledge on the basic concepts of embedded processor*
CLR-2:	*learn the programming techniques in ARM processor*
CLR-3:	*know the steps involved in designing an embedded system*
CLR-4:	*introduce the real time scheduling algorithms*

Course Outcomes (CO):	***At the end of this course, learners will be able to:***	***Programme Outcomes (PO)***		
		1	***2***	***3***
CO-1:	*understand the evolution and architectures of ARM processors*	*1*		
CO-2:	*develop programming skill for ARM processor*	*1*		*2*
CO-3:	*analyze the instruction set and develop program for real world problems*	*1*		*2*
CO-4	*apply the knowledge of embedded system in real time application*	*1*	*1*	*3*

Module-1 - Embedded Processor ***20 Hour***

Review of Embedded Computing; Interfacing sensors embedded system design process; CPS and embedded Computing Architecture of ARM Cortex M3 and Cortex A series processors; ARM tool-chain for compilation, linking and execution phases; Memory system mechanism; Cache; Memory management units and address translation; Performance assessment of embedded processor; Introduction to Embedded Multicore Architecture; ARM virtual hardware: capabilities and use; QEMU for ARM processors

Practice:

1. *Interfacing sensors*
2. *Interfacing LED.*
3. *Interfacing DAC, ADC*
4. *Interfacing DIO, PWMs*

Module-2 - Programming ***18 Hour***

Introduction set, Data transfer, Data processing, conditional and branch instructions, barrier and saturation operations, CortexM4-specific instructions, Thumb2 instructions, Programming of Embedded processors using assembly and C; models for program -- data flow graphs; Assembly language programming of ARM Cortex M3; Hardware software co-design

Practice:

1. *Interfacing keyboard*
2. *Interfacing stepper motor*
3. *Interfacing Temperature sensor*
4. *Interfacing potentiometer sensor*
5. *Interfacing electrical switches*

Module-3 – System Design	***18 Hour***
Design methodologies, Design flows, Requirement analysis, Multiple tasks and multiple processes, Preempt real time operating systems, Priority based scheduling, Distributed embedded systems, Examples of distributed embedded systems in industries, Wireless based embedded systems *Practice:* *1. Interfacing process (time-driven)* *2. Interfacing event schedule* *3. Debugging : Learn to find faults in a system* *4. Tracing : Understand instruction set logs, derive problem loops*	
Module-4 – Applications	***19 Hour***
Processes and real time operating systems; Multi-rate system; real time scheduling algorithms - RMA,EDF and their variants; Energy efficient scheduling algorithms; Data compression techniques, Examples of design of embedded systems. Architecture security features, Arm Confidential Compute Architecture (Arm CCA) – an isolation technology that builds on the strong security foundations of TrustZone- AUTOSAR ECUs with ARM processors : Next generation HPCs and Zonal ECUs *Practice: 1. Communication between MSPs ,2. Networking MSPs using Wi-Fi , 3. Smart automation*	

Learning Resources	1. *Marilyn Wolf,, High Performance embedded Computing: Applications in Cyber Physical Systems and Mobile Computing, 2 nd Edition, Elsevier 2014* 2. *JosephYiu,, The definitive Guide to ARM Cortex M3 and M4 Processors, 3 rd Edition, Elsevier 2020* 3. *Ata Elahi-Trever Arjeski, ―ARM Assembly language with hardware experiment‖, Springer Int. Publishing, 2016.*	4. *William hohl and Christoper Hinds, ―ARM assembly language fundamentals and Techniques ‖CRC, 2 nd edition,2015.* 5. *Marilyn Wolf, Computers as Components: Principles of Embedded Computing System Design, Third Edition, Elsevier 2022* 6. *Daniel Kusswurm, Modern Arm Assembly Language Programming, Apress, 2020*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	*Remember*	20%	-	-	20 %	20%	-
Level 2	*Understand*	20%	-	-	20%	20%	-
Level 3	*Apply*	30%	-	-	30 %	30%	-
Level 4	*Analyze*	30%	-	-	30 %	30%	-
Level 5	*Evaluate*	-	-	-	-	-	-
Level 6	*Create*	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
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